Radiomics Classification Study Report

Radiomics Research Team

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Executive Summary

DECISION: REJECTED

This radiomics study investigated the classification of medical images into 'healthy' versus 'neuroblastoma' categories using texture and shape features extracted from imaging data. The analysis utilized 10 radiomic features including Run Variance, Dependence Non-Uniformity, High Gray Level Zone Emphasis, and morphological characteristics. The machine learning model achieved modest performance with a mean accuracy of 0.60 ± 0.20 across cross-validation folds, though performance metrics showed considerable variability. The evaluation on test samples demonstrated potential discriminative capability with prediction scores ranging from 0.56 to 0.87.

1 Introduction

This research aimed to classify medical imaging data using radiomic features to distinguish between healthy and neuroblastoma cases. Radiomics involves extracting quantitative features from medical images that can reveal disease characteristics not visible to the naked eye.

2 Methodology

2.1 Experimental Design

- **Features**: 10 radiomic features including texture and shape characteristics
- **Target**: Binary classification (healthy vs neuroblastoma)
- **Validation**: 5-fold cross-validation
- **Dataset**: 13 samples with balanced representation

2.2 Feature Extraction

The analysis utilized comprehensive radiomic features including:

- Run Variance
- Dependence Non-Uniformity
- High Gray Level Zone Emphasis
- Morphological characteristics
- Shape-based descriptors

Metric	Mean	Standard Deviation
Accuracy	0.60	± 0.20
AUC	0.40	± 0.20
Recall	0.60	± 0.20
Precision	0.40	± 0.30
F1-score	0.47	± 0.27

Table 1: Model performance metrics across 5-fold cross-validation

3 Results

3.1 Cross-Validation Performance

3.2 Feature Importance

Top discriminative features identified through the modeling process included:

1. Run Variance
2. Dependence Non-Uniformity
3. Surface Volume Ratio
4. Other morphological and texture characteristics

4 Discussion and Conclusion

4.1 Model Performance Assessment

The current model demonstrates insufficient discriminative power for reliable classification between healthy and neuroblastoma cases based on the extracted radiomic features. The performance metrics indicate:

- **Mean accuracy**: 0.60 (below acceptable clinical thresholds)
- **AUC**: 0.40 (indicating poor discriminative capability)
- **High variability**: Significant standard deviations across folds suggest model instability

4.2 Decision Justification

REJECTED - The model performance (mean accuracy: 0.60, AUC: 0.40) is below acceptable thresholds for clinical or research applications, with high variability across folds indicating unreliable predictive capability.

4.3 Recommendations

- Further feature engineering and selection
- Exploration of alternative modeling approaches
- Consideration of additional imaging modalities
- Expansion of dataset size for improved model stability

The current findings suggest that while radiomic features show potential, the current feature set and modeling approach require significant improvement before achieving reliable classification performance.